GCSE MATHEMATICS

Higher Tier Practice Paper 1

Non-Calculator **Time: 1 hour 30 minutes**

Total Marks: 80

Instructions:
- Answer **all** questions.
- Write your answers in the spaces provided.
- You must show all your working. Diagrams are NOT accurately drawn, unless otherwise stated.
- Calculators may NOT be used.

Information:
- The total mark for this paper is 80.
- The marks for each question are shown in brackets [ ].
- Questions are arranged in order of increasing difficulty.

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1. Work out $3\frac{1}{4} + 2\frac{2}{5}$.

2. A map has a scale of 1 : 25,000. The distance between two villages on the map is 7.2 cm. Work out the real distance, in kilometres.

3. (a) Simplify $3x + 5y - 2x + 7y$.

   (b) Solve $5(x - 3) = 2x + 9$.

4. The table shows the number of pets owned by students in a class.

Number of pets	0	1	2	3	4
Frequency	4	9	5	1	1

Work out the mean number of pets.

5. Emma buys a jacket for £84 plus VAT at 20%. She pays with a £100 note. How much change does she receive?

6. (a) Write 84 as a product of its prime factors.

   (b) Find the Highest Common Factor (HCF) of 84 and 120.

7. Here is a sequence: 3, 8, 13, 18, 23, ...

   (a) Write down the next term.

   (b) Find an expression for the nth term.

   (c) Is 101 a term in the sequence? Give a reason for your answer.

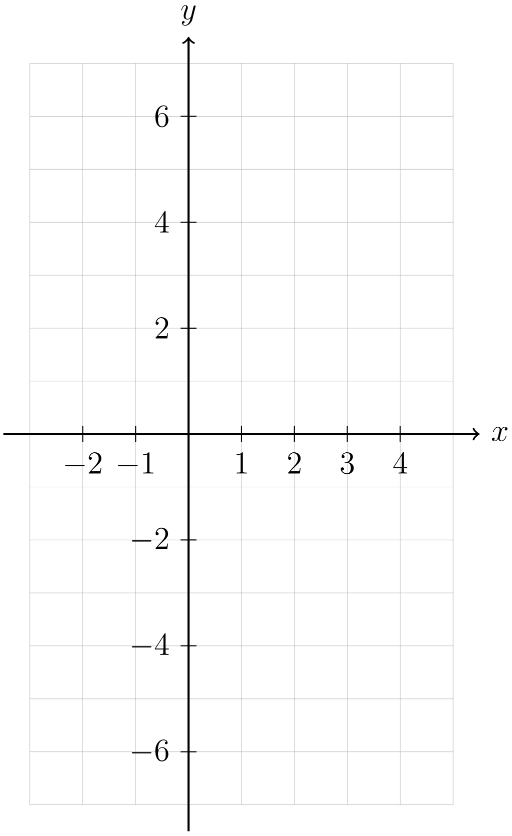
8. A fair 6-sided dice is rolled once.

   (a) What is the probability of rolling a factor of 10?

   (b) The dice is rolled 300 times. How many times would you expect to roll a number greater than 4?

9. Share £450 in the ratio 2 : 3 : 5.

10. On the grid below, draw the graph of $y = x^2 - 2x - 4$ for values of x from -2 to 4.

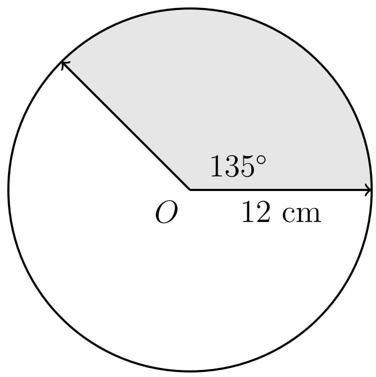


11. (a) Construct triangle PQR where $PQ = 8$ cm, $QR = 6$ cm, and angle $PQR = 60°$. *(Accurately drawn, use a ruler and compasses).*

(b) Measure the length of PR. Give your answer to the nearest millimetre.

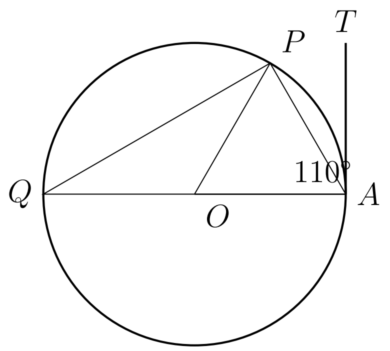
12. A cuboid measures 6 cm by 4 cm by 9 cm. Calculate the length of the space diagonal. Give your answer as a simplified surd.

13. The diagram shows a sector of a circle, centre O, radius 12 cm. The angle of the sector is 135°. Calculate the perimeter of the sector. Give your answer in terms of π.



14. (a) Expand and simplify $(x+5)(x-3)$. (b) Factorise fully $6x^2+13x-5$. (c) Solve $x^2-7x+10=0$.

15. P and Q are points on the circumference of a circle, centre O. AT is a tangent to the circle at A. Angle $OAT = 90°$, angle $AOP = 110°$. Find the size of angle APQ. *Diagram is accurately drawn.*



16. Prove that the sum of the squares of any two consecutive odd numbers is always 2 more than a multiple of 8.

17. The functions f and g are defined as: $f(x) = \frac{3}{x+2}, \quad x \neq -2$ $g(x) = 2x - 1$

    (a) Find $f(4)$.

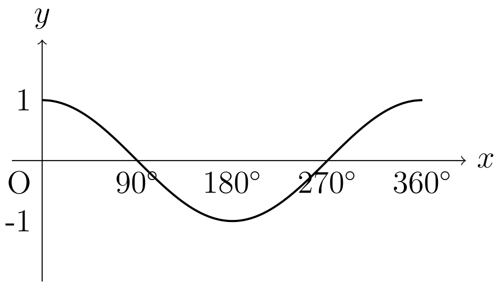
    (b) Find $f^{-1}(x)$.

    (c) Show that $fg(x) = \frac{3}{2x+1}$. State the value of x for which $fg(x)$ is undefined.

18. OAB is a triangle. $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. P is the point on AB such that $AP : PB = 2 : 3$.

   (a) Find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. Give your answer in its simplest form.

   (b) Q is the point on OP such that $OQ : QP = 3 : 1$. Show that AQB is a straight line.



19. The graph of $y = \cos x$ for $0° \leq x \leq 360°$ is shown below.

   (a) On the same axes, sketch the graph of $y = \cos(x - 90°)$ for $0° \leq x \leq 360°$.

   (b) Solve $\cos x = \cos(x - 90°)$ for $0° \leq x \leq 360°$.

20. The iterative formula below can be used to approximate a solution to an equation.

$$x_{n+1} = \frac{5}{x_n - 2} + 2$$

Starting with $x_0 = 4$, work out the values of x_1 and x_2. Give your answers to 1 decimal place where appropriate.

21. Given that $(\sqrt{2} + \sqrt{8})^2 = a + b\sqrt{2}$, find the values of a and b.

END OF PAPER